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The effect of depression on non-suicidal self-injury and psychological status in adolescents with unipolar and bipolar disorders

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Abstract

Objective To investigate the effects of depression on non-suicidal self-injury (NSSI) and related psychological conditions in adolescents with unipolar disorder (UD) and bipolar disorder (BD), and to provide a basis for accurate prevention and intervention of NSSI behaviors in adolescents.

Methods This cross-sectional study collected data from adolescents aged 12–18 years with depressive episodes who exhibited NSSI behaviors and attended the psychiatric outpatient clinic of Huangshi City Psychiatric Specialized Hospital from 2018 to 2023. Depressive episodes were clinically diagnosed by two psychiatrists according to the ICD-10.

Results In terms of NSSI behavioral patterns and severity, adolescents with UD displayed more frequent behaviour of intentionally burning themselves with cigarettes and had more instances of self-inflicted suicidal thoughts that were not carried out compared to those with BD, and the differences between the two groups were statistically significant ($P < 0.05$). For psychological status, Nurses' Global Assessment of Suicide Risk (NGASR) scores were significantly higher in adolescents with BD than in those with UD ($P < 0.05$). There was a significant negative correlation between anxiety scores and the frequency of NSSI in adolescents with BD ($P < 0.05$); that is, the more anxious the adolescents with BD, the lower the frequency of NSSI. The NGASR scores of adolescents with UD and BD were significantly and positively correlated with the frequency of NSSI occurrences ($P < 0.05$); higher NGASR scores of adolescents with UD and BD corresponded with higher frequencies of NSSI.

Conclusion The differences in NSSI behaviors between adolescents with UD and those with BD were statistically significant in terms of self-injury method and severity. Adolescents with UD experienced more severe consequences related to NSSI behaviors. Regarding psychological conditions, adolescents with BD are at a higher risk of suicide. An inverse relationship was observed between anxiety severity and the frequency of NSSI in adolescents with BD; severe anxiety was associated with lower NSSI frequency. Additionally, higher suicide risk was associated with a

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higher frequency of NSSI in adolescent patients with either UD or BD. Therefore, different prevention and intervention measures are needed to address NSSI behaviors in adolescents with UD and BD.

Keywords Unipolar disorder, Bipolar disorder, Adolescent depression, Non-suicidal self-injury, Psychological status

Introduction

Non-suicidal self-injury (NSSI) refers to a series of repeated, intentional, and direct injuries to one's own body tissues, such as cutting and pinching without suicidal intent and lacking social or cultural acceptance of such behavior [1]. NSSI typically manifest in early or mid-adolescence and NSSI among adolescent has become a significant public health and social issue worldwide [2–4]. NSSI behaviors among adolescent are closely related to psychological distress, underscoring the importance of identifying its influencing factors. This is of great practical significance for current effort in adolescent suicide prevention and control, as well as the improvement of adolescent mental health. Existing studies have shown that the occurrence of NSSI behaviors in adolescents involves biological, psychological, and social factors, with mental illness, such as depression and anxiety being the most common influencing factors [5]. In addition, research has shown that depression causes many adverse psychological conditions in adolescents and NSSI behavior is one of the most common strategies used by adolescents to regulate these conditions. Depressive disorders are strongly associated with NSSI, with the severity of depression being positively correlated with NSSI behaviors [6, 7]. Depression and NSSI often co-occur, making depression a major risk factor and predictor of NSSI behaviors [8]. A joint survey of self-injurious behaviors among adolescents across 11 European countries revealed that 27.6% of adolescents had experienced NSSI, with the prevalence rising to 37% among those with depression. In China, the detection rate of NSSI in adolescents ranges from 5.4 to 44.3%, while the rates among depressed adolescents were between 44.0% and 61.2% [9].

Depression is strongly associated with NSSI. However, the differential impact of Unipolar Depression (UD) and Bipolar Disorder (BD) on the development of NSSI, along with their corresponding psychological conditions, has rarely been reported [10–12]. Therefore, this study aimed to investigate NSSI behaviors in adolescents with depression during UD and BD, explore the differences in NSSI behaviors and psychological conditions between these two groups and provide a theoretical basis for psychotherapeutic interventions.

Case analysis and methods

Case collection and inclusion criteria

Patients attending outpatient clinics at Huangshi Hospital for Mental Diseases between October 2018 and

October 2023 were included in the study based on the following criteria: [1] Adolescents aged 12–18 years [2]. Clinical diagnosis of depressive episodes by at least two psychiatrists based on the ICD-10, including adolescents with UD and BD [3]. Prior NSSI Behavior. Exclusion criteria: [1] intellectual disability [2]. Refusal to sign the informed consent. A total of 545 valid research participants were included in this study, and the patients were categorized into two groups: UD and BD. Among them, 361 adolescents with UD and 184 adolescents with BD were enrolled (mean age: 15 years; 163 males, 382 females; adolescents with UD: 110 males, 251 females; adolescents with BD: 53 males, 131 females). The study was approved by the Ethics Committee of Huangshi Psychiatric Hospital (HSMHC20231201), and written informed consent was obtained from all participants or their guardians.

Methods

This cross-sectional survey employed a random sampling technique. Specifically, the surveys utilized the following tools: the Hamilton Anxiety Scale (HAMA) [13], Hamilton Depression Scale (HAMD) [14, 15], Symptom Checklist-90 (SCL-90) [16], Deliberate Self-Harm Inventory (DSHI) [17], Nurses' Global Assessment of Suicide Risk (NGASR) [18], and four psychological condition scales related to NSSI behavior. All the scales were the validated Chinese versions with good reliability and validity.

General situation

Data collected included sex, age, family economy status, place of residence, and whether the participant was an only child.

HAMA

HAMA was used to assess the anxiety level of the patients with 14 items [13]. All items were scored on a 5-point scale from 0 to 4, and the total score was the sum of all items, i.e., 0 to 56. A higher total score suggests more severe anxiety symptoms [19, 20]. The scale Cronbach's alpha coefficient was 0.89 [19]. A score of less than 14 on the HAMA scale suggests that the patient does not have an anxiety disorder [20]. If the HAMA scale is ≥ 14 , a differential diagnosis of the type of anxiety is made, which includes Generalized Anxiety Disorder, Panic Disorder, Plaza Fear Disorder, Social Anxiety Disorder, Dissociative Anxiety Disorder, Specific Fear Disorder, Unspecified Fear Disorder, and Selective Mutism Anxiety Disorder. In this study, patient-related anxiety types identified included only Generalized Anxiety Disorder, Panic

Disorder, Social Anxiety Disorder, and Unspecified Fear Disorder [20].

HAMD

HAMD is an examiner-rating scale to assess the level of depression in patients and 24-item version was applied in this study [14]. All items were rated on a 5-point scale from 0 to 4, and the total score was the sum of all items, i.e., 0 to 96. The scale Cronbach's alpha coefficient was 0.88 [14]. The HAMD assessment method is simple, with clear and easy-to-understand standards [21]. It can be used to evaluate depressive symptoms, such as depression, BD, neurosis, and other diseases [22]. The HAMD is particularly effective in reflecting the severity of depressive symptoms; the higher the total score, the more severe the condition [22, 23].

SCL-90(Symptom check List-90)

The SCL90 was used to assess the psychological status of patients [16]. The SCL-90 consists of nine factors: somatization, obsessive-compulsive symptoms, interpersonal sensitivity, depression, anxiety, hostility, phobia, paranoia, and psychoticism [24]. Each factor reflects a particular aspect of an individual's symptom characteristics, and factor scores provide insights into the distribution of symptoms [25]. Each dimension is scored from 1 to 5, with higher scores indicating more severe symptoms and poorer psychological status [26].

DSHI

The DSHI uses the Deliberate Self-Injury Scale developed by Kim, a behavior-based self-report questionnaire that records the types and frequencies of NSSI behaviors over the past years [27]. This scale provides a detailed assessment of the NSSI modalities, including striking, burning, and carving [17, 27]. In this study, the DSHI scale was supplemented to include additional modalities of self-injury not listed above, which were categorized as either lethal self-injury or non-lethal self-injury based on keywords and previous literature [17, 28]. The scale showed good reliability in this study [28].

NGASR

The NGASR is an examiner-rating scale for assessing suicide risk [18]. It is a simple, quick, accurate and effective way to screen patients at clinical risk of suicide so that targeted interventions can be given to prevent unintentional events [29]. Psychiatrists used the NGASR for initial diagnosis, and it consists of 15 items: feelings of hopelessness, recent negative life events, delusions or hallucinations of victimization, depressed mood or loss of interest or pleasure, withdrawal from interpersonal and social functioning, verbal disclosure of suicidal ideation, planning of suicidal actions, family history of

suicide, death of a close family member or loss of a significant intimate relationship, history of psychiatric illness, widowhood or widowerhood, history of suicide attempts, low socioeconomic status, history of alcohol use or abuse, and history of advanced illness [30]. These items indicate a higher risk of suicide among adolescents [31, 32]. Differences in suicide risk among adolescents with UD or BD were assessed using this scale. This scale is reliable for research [31, 32].

Statistical analyses

SPSS version 29.0 was used for statistical analysis. All continuous variables were first tested for normality. Variables conforming to a normal distribution were described using $\bar{x} \pm S$, while non-normally distributed variables were described using quartiles. To compare the differences in characteristics between the two groups, analysis of variance (ANOVA) was employed for continuous variables, the χ^2 test for categorical variables, and Mann-Whitney U test was utilized for non-normally distributed variables.

Results

General information

In this study, 545 adolescents with depressive episodes and previous self-injurious behaviors were examined. These participants were categorized into two groups for analysis: 361 adolescents with UD and 184 adolescents with BD. The majority of the cases were female (70%), students (96%), Non-only-children (68%) and dwelling in the city (70%). It is worth mentioning that 52% of the patients have no complete self-awareness of their diseases, 46% of the patients are living under the pressure of study or work, and 54% of the patients feel that they are not understood by their families. No significant differences were found in age, gender, family structure, whether they were only child, family situation, level of education, presence of past history, self-awareness status, dwelling area, mental illness family pedigree among patients with BD and UD groups. Patients in the BD group have a longer disease duration than those in the UD group ($p < 0.05$). (Table 1).

Symptom clusters of depression

Depression with anxiety disorder

The results of this study, as presented in Table 2 revealed that approximately 47% of the patients had a HAMA Scale score greater than or equal to 14, it means that this part of patients suffers from an anxiety disorder. In addition, no significant difference was observed between adolescents with and without anxiety disorders, regardless of whether they had UD or BD (Fisher, $P = 0.294$). However, among the adolescent patients with UD and BD who suffered from anxiety disorders, generalized anxiety

Table 1 Basic characteristics of adolescents with UD and BD

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	z/χ^2	<i>p</i>
Sex, n (%)				0.092	0.762
Male	163 (30)	110 (30)	53 (29)		
Female	382 (70)	251 (70)	131 (71)		
Age, Median (Q1, Q3)	15(14,17)	15(14,17)	15(14,17)	1.264	0.206
Occupation, n (%)				5.678	0.017
Student	521 (96)	351 (97)	170 (92)		
Staying at home	24 (4)	10 (3)	14 (8)		
Only child, n (%)				0.398	0.528
Only child	174 (32)	119 (33)	55 (30)		
Non-only-child	371 (68)	242 (67)	129 (70)		
Place of residence, n (%)				0.382	0.536
Urban	381 (70)	256 (71)	125 (68)		
Rural	164 (30)	105 (29)	59 (32)		
Family structure, n (%)				Fisher	0.132
Nuclear family	381 (70)	258 (72)	123 (67)		
Host family	10 (2)	4 (1)	6 (3)		
Left-behind families	80 (15)	47 (13)	33 (18)		
Single parent families	67 (12)	48 (13)	19 (10)		
Reorganized family	7 (1)	4 (1)	3 (2)		
Family situation, n (%)				0.047	0.829
Family of origin	455 (83)	300 (83)	155 (84)		
Non-origin family	90 (17)	61 (17)	29 (16)		
Education level, n (%)				1.762	0.623
Elementary school	18 (3)	11 (3)	7 (4)		
Middle school	206 (38)	140 (39)	66 (36)		
High school (junior college/technical school)	290 (53)	187 (52)	103 (56)		
University and above (college)	31 (6)	23 (6)	8 (4)		
Disease duration, n (%)				14.895	0.005
≤ half a year	177 (33)	131 (36)	46 (25)		
Half a year < duration ≤ one year	143 (26)	101 (28)	42 (23)		
One year < duration ≤ two years	89 (16)	51 (14)	38 (21)		
Two years < duration ≤ four years	97 (18)	58 (16)	39 (21)		
More than four years	39 (7)	20 (6)	19 (10)		
Past history, n (%)				0.686	0.407
No	507 (93)	333 (92)	174 (95)		
Yes	38 (7)	28 (8)	10 (5)		
Mental illness family pedigree, n (%)				0.833	0.361
No	504 (92)	337 (93)	167 (91)		
Yes	41 (8)	24 (7)	17 (9)		
School or workplace violence, n (%)				22.482	< 0.001
No	160 (29)	83 (23)	77 (42)		
Yes	41 (8)	26 (7)	15 (8)		
Unknown	344 (63)	252 (70)	92 (50)		
Stress at school or work, n (%), n (%)				12.994	0.002
No	71 (13)	39 (11)	32 (17)		
Yes	250 (46)	155 (43)	95 (52)		
Unknown	224 (41)	167 (46)	57 (31)		
Self-awareness, n (%)				0.804	0.370
Complete self-awareness	259 (48)	177 (49)	82 (45)		
No complete self-knowledge	286 (52)	184 (51)	102 (55)		
Family atmosphere, n(%)				Fisher	0.002
Harmony	26 (5)	16 (4)	10 (5)		

Table 1 (continued)

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	z/χ^2	<i>p</i>
Perceived lack of understanding by family	292 (54)	190 (53)	102 (55)		
Parental pressure	57 (10)	27 (7)	30 (16)		
Domestic violence	3 (1)	3 (1)	0 (0)		
Other	167 (31)	125 (35)	42 (23)		

Abbreviations UD: unipolar disorder; BD: bipolar disorder

Table 2 Adolescents with UD and BD exhibiting anxiety disorders

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	z/χ^2	<i>p</i>
Differential Diagnosis Anxiety Type, n (%)				Fisher	0.294
No anxiety	291 (54)	202 (56)	89 (48)		
Generalized anxiety disorder	236 (43)	147 (41)	89 (48)		
Panic disorder	1 (0)	1 (0)	0 (0)		
Social anxiety disorder	17 (3)	11 (3)	6 (3)		

Abbreviations UD: unipolar disorder; BD: bipolar disorder; SCL-90: Symptom Check List-90

disorder was the most common (43%), followed by social anxiety disorder (3%). Notably, only one participant suffered from panic disorder and unspecified fear disorder (0%).

SCL-90 assessment

We conducted 465 SCL-90 assessments in 317 adolescents with UD and 148 adolescents with BD. Of these, 391 (84%) exhibited psychiatric symptom cluster on the SCL-90, while 74 (16%) did not, and there was no statistically significant difference between the BD and UD groups ($\chi^2=1.891$, $P=0.169$). Specifically, the SCL-90 results showed that patients in the BD group had more severe somatic, obsessive-compulsive, anxiety, paranoia, psychotic symptoms and greater interpersonal sensitivity than those in the UD group. In addition, the difference was not statistical significant between adolescents with UD and those with BD in the presence of concomitant depressive symptoms, hostility, and phobias anxiety. (Table 3).

Sleep disorder

The study results presented in Table 4 revealed that out of the 545 data points collected on the sleep status of adolescent patients, the vast majority of patients exhibited sleep disorders (91%), only 9% patients did not. Among these patients, 85% had difficulty falling asleep, 49% experienced early awakening, 33% suffered from intermittent light sleep, 6% had excessive sleepiness, and 3% had a decreased need for sleep. It should be noted that the results revealed patients with BD were more likely to

Table 3 SCL-90 assessment in adolescents with UD and those with BD

Variables	Total (n = 465)	UD (n = 317)	BD (n = 148)	z/χ^2	<i>p</i>
SCL-90, n (%)				1.891	0.169
Asymptomatic	74 (16)	56 (18)	18 (12)		
Symptomatic	391 (84)	261 (82)	130 (88)		
Somatization				12.430	<0.001
No	316(68)	223(74)	83(56)		
Yes	149(32)	84(26)	65(44)		
Obsessive-compulsive, n (%)				5.910	0.015
No	203 (44)	151 (48)	52 (35)		
Yes	262 (56)	166 (52)	96 (65)		
Interpersonal sensitivity, n (%)				4.615	0.032
No	313 (67)	224 (71)	89 (60)		
Yes	152 (33)	93 (29)	59 (40)		
Depression, n (%)				3.383	0.066
No	148 (32)	110 (35)	38 (26)		
Yes	317 (68)	207 (65)	110 (74)		
Anxiety, n (%)				9.253	0.002
No	177 (38)	136 (43)	41 (28)		
Yes	288 (62)	181 (57)	107 (72)		
Hostility, n (%)				0.505	0.477
No	239 (51)	167 (53)	72 (49)		
Yes	226 (49)	150 (47)	76 (51)		
Phobic anxiety, n (%)				1.688	0.194
No	313 (67)	220 (69)	93 (63)		
Yes	152 (33)	97 (31)	55 (37)		
Paranoid ideation, n (%)				6.724	0.010
No	231 (50)	171 (54)	60 (41)		
Yes	234 (50)	146 (46)	88 (59)		
Psychoticism, n (%)				4.630	0.031
No	289 (62)	208 (66)	81 (55)		
Yes	176 (38)	109 (34)	67 (45)		

Abbreviations UD: unipolar disorder; BD: bipolar disorder; SCL-90: Symptom Check List-90

experience intermittent light sleep and decrease sleep - demand than those with UD. In terms of other types of sleep disorders, the differences between the two groups were not statistically significant.

Table 4 Sleep disorder among adolescents with UD and BD

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	z/χ^2	<i>p</i>
Have a sleep disorder, n (%)				0.297	0.586
No	48 (9)	34 (9)	14 (8)		
Yes	497 (91)	327 (91)	170 (92)		
Classification of sleep disorders					
Difficulty falling asleep, n (%)				0.769	0.381
No	83 (15)	51 (14)	32 (17)		
Yes	462 (85)	310 (86)	152 (83)		
Early awakening, n (%)				2.679	0.102
No	280 (51)	195 (54)	85 (46)		
Yes	265 (49)	166 (46)	99 (54)		
Intermittent light sleep, n (%)				8.445	0.004
No	366 (67)	258 (71)	108 (59)		
Yes	179 (33)	103 (29)	76 (41)		
Hypersomnia, n (%)				0.164	0.686
No	514 (94)	342 (95)	172 (93)		
Yes	31 (6)	19 (5)	12 (7)		
Reduced need for sleep, n (%)				14.156	< 0.001
No	527 (97)	357 (99)	170 (92)		
Yes	18 (3)	4 (1)	14 (8)		

Abbreviations UD: unipolar disorder; BD: bipolar disorder

Somatic and psychological symptom clusters of depression

Based on the records of the diagnoses and somatic symptoms of the patients by two psychiatrists, among the 545 adolescents with UD or BD, 97% of the patients had obvious somatic symptoms. It is worth noting that 91% of the patients suffer from sleep disorders, 73% of the patients have eating and weight disorders, and 81% of the patients experience energy loss. Compared with the BD group, patients in the UD group had more severe energy loss ($p=0.003<0.005$). There were no significant statistical differences in the other somatic symptoms between the two groups of patients. (Table 5)

Similarly, according to the diagnoses of the patients and the statistics of psychological symptom groups by two psychiatrists, the results in Table 6 show that patients in the UD group have more severe anxiety ($p<0.001$), thought retardation ($p=0.022<0.05$) and psychomotor retardation ($p<0.001$), while patients in the BD group have more severe psychotic symptoms ($p<0.001$) and suicidal and self-harming thoughts ($p=0.037<0.05$).

Differences in the style and severity of NSSI behavior

Of the 17 modes of self-injury listed on the DSHI scale, along with the two additional modes of self-injury (Table 7), there were 187 (34%) patients deliberately

cutting their wrists, arms or other body areas, 141 patients (26%) deliberately burning themselves with cigarettes and 78 patients (14%) having suicidal thoughts but not carrying them out. Compared with the patients in the BD group, those in the UD group had more behaviors of deliberately burning themselves with cigarettes ($p<0.001$), while the BD group had more suicidal thoughts ($p=0.018<0.05$). In other NSSI behaviors, the numbers of patients in both groups were relatively small and there were no statistically significant differences.

Differences in psychological status

As presented in Table 8, there was no statistically significant difference in the HAMA and HAMD scores between the UD and BD groups ($P>0.05$). However, the BD group had a significantly higher risk of suicide compared to the UD group, which was statistically different between the two groups ($t=-2.66$, $P=0.008$).

Correlation analysis between psychological status and NSSI frequency

In this study, anxiety and NSSI frequency were not significantly correlated in the UD group. In contrast, anxiety and NSSI frequency showed a significant negative correlation in the BD group ($P<0.05$); that is, worse anxiety was associated with lower NSSI frequency. Depression and NSSI frequency were not significantly correlated in either the UD or PD groups ($P>0.05$). Suicide risk and NSSI frequency were significantly positively correlated in both the UD and BD ($P<0.05$); that is, the higher the suicide risk, the higher the NSSI frequency (Table 9).

Discussion

NSSI is a widespread public health problem worldwide, and its onset is closely related to family, neurobiological, and social factors that significantly affect the physical and mental health of adolescents [33–35]. Currently, there is no unified program for the prevention and treatment of NSSI, and most interventions are pharmacological and psychotherapeutic. Although research on NSSI has advanced, many issues still require further exploration [36, 37]. In the present study, there were no significant differences between adolescents with UD and adolescents with BD in terms of gender, age, occupation, whether they were only child, place of residence, family structure, family situation, level of education, presence of past history, mental illness family pedigree and self-awareness status. Most of the literature found no statistically significant differences between patients with UD and those with BD when comparing information on years of education, age of first onset of disease, age of first consultation, age of diagnosis, gender, and having a family history of the disease [38, 39]. Some scholars in their

Table 5 Somatic symptom profiles of adolescents with UD and BD

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	z/χ^2	p
Presence of significant physical symptoms, n (%)				1.508	0.219
No	527 (97)	352 (98)	175 (95)		
Yes	18 (3)	9 (2)	9 (5)		
Classification of symptoms					
Sleep disorder, n (%)				0.297	0.586
No	48 (9)	34 (9)	14 (8)		
Yes	497 (91)	327 (91)	170 (92)		
Eating and weight disorders, n (%)				0.595	0.441
No	147 (27)	93 (26)	54 (29)		
Yes	397 (73)	267 (74)	130 (71)		
Loss of energy, n (%)				8.669	0.003
No	103 (19)	55 (15)	48 (26)		
Yes	442 (81)	306 (85)	136 (74)		
Depression is heavy at daytime and light at night, n (%)				0.077	0.781
No	524 (96)	346 (96)	178 (97)		
Yes	21 (4)	15 (4)	6 (3)		
Depression is light at day and heavy at night, n (%)				1.465	0.226
No	437 (80)	295 (82)	142 (77)		
Yes	107 (20)	65 (18)	42 (23)		
Sexual dysfunction, n (%)				Fisher	1.000
No	545 (100)	361 (100)	184 (100)		
Pain in any part of the body such as headache and backache, n (%)				0.510	0.475
No	480 (88)	321 (89)	159 (86)		
Yes	65 (12)	40 (11)	25 (14)		
Dry mouth and sweating, n (%)				Fisher	0.342
No	540 (99)	359 (99)	181 (98)		
Yes	5 (1)	2 (1)	3 (2)		
Panic and chest tightness, n (%)				0.375	0.540
No	459 (84)	307 (85)	152 (83)		
Yes	86 (16)	54 (15)	32 (17)		
Nausea, vomiting, burning sensation in stomach, n (%)				0.476	0.490
No	516 (95)	344 (95)	172 (93)		
Yes	29 (5)	17 (5)	12 (7)		
Urinary frequency and urgency, n (%)				Fisher	1.000
No	545 (100)	361 (100)	184 (100)		

Table 5 (continued)

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	z/χ^2	p
Blurred vision, n (%)				Fisher	1.000
No	545 (100)	361 (100)	184 (100)		

Abbreviations UD: unipolar disorder; BD: bipolar disorder

research also proposed that age is a relevant factor for patients with first onset of depression and patients with first onset of mania, arguing that patients with first onset of depression themselves have the problem of functional disorder of thinking, and along with the passage of time the negative emotions generated by self-denial become more and more serious, and the sense of somatic maladjustment increases the feeling of anxiety, so that in the case of decreasing organismic function depression symptoms will be more pronounced [40, 41]. This is different from the results of this study, or related to the number of samples and the lack of representativeness of the samples.

In this study, Both groups possibly experienced anxiety symptoms, with generalized anxiety disorder being predominant, with panic disorder and social anxiety disorder being rare. No differences were seen between the two groups of adolescent patients in terms of concomitant anxiety characteristics, suggesting that depression with anxiety is a more common clinical phenomenon [42].

Compared to adolescents with UD, those with BD exhibited more pronounced somatization, obsessive-compulsive symptoms, greater interpersonal sensitivity, more severe anxiety, paranoid and psychotic symptoms according to the results of the SCL-90 in this study. There were no statistical differences between adolescents with UD and those with BD in the presence of concomitant depressive symptoms, hostility, and phobias anxiety. Previous studies have found that patients with BD have higher rates of obsessive-compulsive, panic attacks, and psychotic symptoms, more bad habits, more previous episodes, and a high percentage of family history [43–45]. It may be that BD symptoms are not only associated with high and low moods, but also with severe impairments in autonomy, cognitive functioning, social skills, spare time management and general functioning [43–45].

In terms of sleep disturbances, both groups experienced some sleep disturbances, but intermittent light sleep was more pronounced among adolescents with UD, while decreased sleep demand was more common among adolescents with BD. In contrast, there was no statistical difference between the two groups of adolescents in terms of difficulty falling asleep, early awakening, and sleepiness. In previous studies, Yates et al. [46] reported that about 3/4 of depressed patients had difficulty falling asleep or maintaining sleep. And

Table 6 Psychological symptom clusters in depressed adolescents with UD and with BD

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	z/χ^2	<i>p</i>
No apparent abnormality, n (%)				Fisher	0.067
No	9 (2)	3 (1)	6 (3)		
Yes	536 (98)	358 (99)	178 (97)		
Symptom classification					
Anxiety, n (%)				14.663	< 0.001
No	111 (20)	56 (16)	55 (30)		
Yes	434 (80)	305 (84)	129 (70)		
Self-blame and self-guilt, n (%)				1.008	0.315
No	90 (17)	55 (15)	35 (19)		
Yes	455 (83)	306 (85)	149 (81)		
Slow thinking, n (%)				5.268	0.022
No	52 (10)	27 (7)	25 (14)		
Yes	493 (90)	334 (93)	159 (86)		
Cognitive symptoms, n (%)				3.405	0.065
No	33 (6)	17 (5)	16 (9)		
Yes	512 (94)	344 (95)	168 (91)		
Psychomotor retardation, n (%)				36.008	< 0.001
No	222 (41)	114 (32)	108 (59)		
Yes	323 (59)	247 (68)	76 (41)		
Psychomotor agitation, n (%)				Fisher	1.000
No	545 (100)	361 (100)	184 (100)		
Psychotic symptoms, n (%)				12.685	< 0.001
No	418 (77)	294 (81)	124 (67)		
Yes	127 (23)	67 (19)	60 (33)		
Suicide and self-injury, n (%)				4.333	0.037
No	351 (64)	244 (68)	107 (58)		
Yes	194 (36)	117 (32)	77 (42)		

Abbreviations UD: unipolar disorder; BD: bipolar disorder

Riemann et al. [47] concluded that insomnia was not only a typical symptom of depression, but also an independent risk factor for predicting the onset of depression through the systematic follow-up of insomnia symptoms in depressive disorders. Sleep disorders are also very common in bipolar disorder, and it has been reported that throughout the course of BD, 70–99% of patients experience sleep problems such as difficulty in falling asleep, shortened sleep duration, and difficulty in maintaining sleep at some time of the disease, and that insomnia is more prevalent, especially during depressive episodes [48, 49].

In the study, adolescents with UD often experience significant energy loss in terms of somatic symptoms. They also frequently presented with slow thinking and psychomotor retardation, adolescents with BD had more severe psychotic symptoms and NSSI thoughts. There were no statistical differences between the two groups of adolescents in terms of self-blame, cognitive symptoms, and psychomotor agitation. It is worth noting that in the statistics of psychological symptom groups by the two doctors, the anxiety of patients in the UD group was more severe than that in the BD group. However, the results of the SCL-90 scale showed that the anxiety of patients in the BD group was more severe than that in the UD group. The possible reason for these two opposite results may be that the perspectives of doctor evaluation and patient self-evaluation are different. Clinically, psychotic symptoms, psychomotor retardation, and agitation were more common in patients with BD; patients with a longer history of BD were also more likely to have co-morbid anxiety disorders and somatic disorders [50–52]. Notably, adolescents with UD often engaged in intentional self-injury, such as burning themselves with cigarettes and BD had more self-injurious suicidal thoughts that were not acted upon. In terms of psychological status, adolescents with BD scored higher on the Suicide Risk Factor Assessment. They had a negative correlation between the Anxiety Rating Scale and the NSSI; that is, higher anxiety score were associated with lower NSSI frequency. Both the UD and BD groups showed significant positive correlation between Suicide Risk Factor Assessment Scale scores and NSSI frequency; that is, higher suicide risk was associated with higher NSSI frequency.

Adolescents have a high incidence of NSSI, especially among those with depression, and the high comorbidity of depression and anxiety indicates a higher sensitivity to stress, such as school or work stress, which, in turn, affects how NSSI occurs and the severity of self-injurious behavior [53, 54]. One foreign study showed that anxiety is significantly related to NSSI behavior; higher levels of anxiety increase the probability of NSSI, and individuals may engage in more severe self-injury to regulate their bad mood and relieve anxiety [55]. However, in this study, there was no significant relationship between anxiety and the occurrence of NSSI in adolescent patients with UD. Conversely, a significant relationship between anxiety and the occurrence of NSSI was observed among adolescent patients with BD, but higher levels of anxiety were associated with lower frequency of NSSI. In addition, Plener et al. [56] highlighted the importance of alleviating depression in interventions for adolescents with NSSI. In the present study, adolescents with UD and BD were more

Table 7 Differences in NSSI behavioral styles and severity between adolescents with UD and BD

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	χ^2	p
Intentional slitting of wrists, arms or other areas of the body, n (%)				0.694	0.405
No	358 (66)	242 (67)	116 (63)		
Yes	187 (34)	119 (33)	68 (37)		
Intentionally burned yourself with a cigarette, n (%)				11.095	< 0.001
No	404 (74)	251 (70)	153 (83)		
Yes	141 (26)	110 (30)	31 (17)		
Intentionally burned yourself with a lighter or match, n (%)				Fisher	1
Yes	544 (100)	360 (100)	184 (100)		
Yes	1 (0)	1 (0)	0 (0)		
Intentionally carving into skin, n (%)				Fisher	1
No	545 (100)	361 (100)	184 (100)		
Intentionally carving designs on the skin, n (%)				Fisher	0.552
No	543 (100)	359 (99)	184 (100)		
Yes	2 (0)	2 (1)	0 (0)		
Deliberately biting yourself on the skin, n (%)				Fisher	1
No	545 (100)	361 (100)	184 (100)		
Deliberate rubbing of sandpaper on body, n (%)				Fisher	1
No	545 (100)	361 (100)	184 (100)		
Deliberately dripping strong acid on skin, n (%)				Fisher	1
No	545 (100)	361 (100)	184 (100)		
Intentionally scrubbing the skin with bleach, n (%)				Fisher	1
No	545 (100)	361 (100)	184 (100)		
Sticking sharp objects into the skin, n (%)				Fisher	1
No	545 (100)	361 (100)	184 (100)		
Scratching skin with glass, n (%)				Fisher	1
No	544 (100)	360 (100)	184 (100)		
Yes	1 (0)	1 (0)	0 (0)		
Destroying or breaking your own bones, n (%)				Fisher	1
No	545 (100)	361 (100)	184 (100)		
Banging head against something, n (%)				Fisher	0.432
No	538 (99)	355 (98)	183 (99)		
Yes	7 (1)	6 (2)	1 (1)		
Hitting hard with your fist, n (%)				Fisher	1
No	540 (99)	358 (99)	182 (99)		
Yes	5 (1)	3 (1)	2 (1)		
Intentionally prevent the wound from coming back, n (%)				Fisher	1
No	542(99)	359(99)	183(99)		
Yes	3(1)	2(1)	1(1)		
Pounding the wall with the fist, n (%)				Fisher	1
No	542 (99)	359 (99)	183 (99)		
Yes	3 (1)	2 (1)	1 (1)		
Pulling your hair hard, n (%)				Fisher	0.114
No	543 (100)	361 (100)	182 (99)		
Yes	2 (0)	0 (0)	2 (1)		
Thought about but did not commit suicide by self-injury, n (%)				5.621	0.018
No	467 (86)	319 (88)	148 (80)		
Yes	78 (14)	42 (12)	36 (20)		
Overdose, n (%)				Fisher	0.452
No	537 (99)	357 (99)	180 (98)		
Yes	8 (1)	4 (1)	4 (2)		

Abbreviations UD: unipolar disorder; BD: bipolar disorder

Table 8 Comparison of psychological status of adolescents with UD and BD ($\bar{X} \pm S$)

Variables	Total (n = 545)	UD (n = 361)	BD (n = 184)	t	P
HAMA Score	15.93 ± 7.67	15.93 ± 7.70	15.93 ± 7.62	-0.01	0.995
HAMD Score	26.42 ± 9.07	26.75 ± 9.01	25.76 ± 9.17	1.21	0.229
NGASR Score	8.41 ± 3.05	8.15 ± 2.84	8.92 ± 3.38	-2.66	0.008

Abbreviations UD: unipolar disorder; BD: bipolar disorder; HAMA: Hamilton Anxiety Scale; HAMD: Hamilton Depression Scale

Table 9 Analysis of the correlation between psychological status and frequency of NSSI in adolescents with UD and BD

Variables	NSSI frequency			
	UD (n = 361)		BD (n = 184)	
	r ^s	p	r ^s	p
HAMA Score	0.039	0.458	-0.186*	0.011
HAMD Score	0.046	0.382	-0.075	0.314
NGASR score	0.170**	0.001	0.167*	0.024

(Note: Spearman correlation analysis ** $p < 0.01$, * $p < 0.05$). Abbreviations: UD: unipolar disorder; BD: bipolar disorder; HAMA: Hamilton Anxiety Scale; HAMD: Hamilton Depression Scale; NGASR: Nurses' Global Assessment of Suicide Risk; NSSI: non-suicidal self-injury

likely to engage in “intentional cigarette burning” compared to those with BD alone, resulting in more severe and lethal consequences. Moreover, adolescents with BD may be more sensitive to stress and experience more severe negative emotions compared to those with UD. This increased sensitivity may lead them to engage in more severe NSSI behaviors as a way to cope. The increasing frequency of NSSI behaviors may indicate that their ability to regulate negative emotions is gradually failing, leading to stronger suicidal ideation when faced with strong negative emotions. Therefore, more attention should be paid to NSSI in both adolescents with UD and those with BD, as both conditions are of great significance for the prevention and control of suicide among adolescents.

Several studies have indicated that the prevalence of NSSI is higher among non-only-child children compared to that of the only-child children [57, 58]. National and international studies have also found statistically significant sex differences in NSSI behaviors, with females experiencing a higher frequency of NSSI. This may be because females use different emotional regulation strategies than males to cope with their own stresses and are more prone to triggering undesirable emotions, such as depression or anxiety, which may lead to more self-injurious behaviors [59]. Similarly, in the present study, adolescent females with either UD or BD were predominant and the majority were non-only-child children, residing in urban areas.

The occurrence of NSSI in adolescents results from the interaction of multiple factors. The psychological condition is closely related to NSSI, and the lack

of social support and emotional communication are significant risk factors for NSSI [60, 61]. Studies on adolescents with NSSI and depressive disorders have found that those with depressive disorders and NSSI who lack effective social communication may experience high levels of perceived stress and state anxiety [62, 63]. The present study further explored the differences in psychological conditions related to NSSI behaviors in the groups of adolescent patients with UD and BD. The result revealed a statistically significant difference in suicide risk between the UD and BD groups. Adolescents with BD are more likely to react negatively and are potentially at higher risk when faced with school or work stress. This may be because adolescents with NSSI are more prone to negative emotions, such as anxiety, after being exposed to adverse stressful stimuli, and high anxiety levels can lead to greater sensitivity, which makes adolescents more prone to impulsively engage in NSSI behaviors.

Adolescence is an important phase in life characterized by rapid physical and mental development. However, it is also a time when adolescents face various pressures; if these pressures are not handled properly, they can result in emotional disorders, such as depression, anxiety, self-injurious behaviors, and other adverse psychological conditions [64, 65]. Therefore, families, schools, and society should actively guide adolescents to adopt effective strategies to manage their emotions. This is especially crucial for adolescents experiencing depression or anxiety, who should receive timely treatment. In addition, it is recommended that adolescents with UD or BD seek help from professional counselors or medical personnel to prevent and address their NSSI behaviors.

Limitations

This study has several shortcomings; first, the information on the assessment of SCL90 symptoms was flawed. Second, this study only discusses the effect of NSSI in adolescent patients with UD and BD without incorporating a control group of normal adolescents for comparison. Therefore, further investigation is needed to increase the sample size and conduct a multicenter study to examine the effect of supplementation thoroughly.

Conclusion

The differences in NSSI behaviors between adolescents with UD and those with BD were statistically significant in terms of self-injury method and severity. Adolescents with UD experienced more severe consequences related to NSSI behaviors. Regarding psychological conditions, adolescents with BD are at a higher risk of suicide. An inverse relationship was observed between anxiety severity and the frequency of NSSI in adolescents with BD; severe anxiety was associated with lower NSSI frequency.

Additionally, higher suicide risk was associated with a higher frequency of NSSI in adolescent patients with either UD or BD. Therefore, different prevention and intervention measures are needed to address NSSI behaviors in adolescents with UD and BD.

Abbreviations

UD	Unipolar disorder
BD	Bipolar disorder
SCL-90	Symptom Check List-90

Acknowledgements

We would like to thank Editage (<http://www.editage.cn>) for English language editing.

Author contributions

Zhuofan Ye and Fanshi Zhang conceived and wrote the first draft, Ruxue Cui, Xixiang Ye, and Mengqing Tan collected and analyzed the data for this article, Tao Tao and Xiaozhi Zhang supervised the writing of this article. All authors have read and agreed to the published version of the manuscript.

Funding

None.

Data availability

The data that support the findings of this study are available from the corresponding author, [Xiaozhi Zhang], upon reasonable request.

Declarations

Ethics approval and consent to participate

The procedures used in this study adhere to the tenets of the Declaration of Helsinki. This study was approved by the Ethics Committee of Huangshi Psychiatric Hospital (Grant No.: HSMHC20231201). Informed consent was obtained from all participants in this study. For minors, we have obtained informed consent from their guardians.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

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Received: 10 October 2024 / Accepted: 27 November 2024

Published online: 18 December 2024

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